# ABDULLAH SARWAR

*Senior Full-Stack Engineer · React.js · Vue.js · React Native · AI Integration*

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## PROFESSIONAL SUMMARY

I'm a Senior Full-Stack Engineer with over 7 years of experience building scalable web and mobile applications — from early-stage MVP launches to platforms serving 200M+ users. My work spans the full product lifecycle: discovery, architecture, design, development, and deployment. I genuinely enjoy untangling complex problems and turning them into clean, performant solutions.

Beyond the traditional stack, I've integrated AI models — both pre-trained (OpenAI, Fireworks AI) and custom-trained — to create more intelligent, human-like product experiences. I thrive in Agile, cross-functional teams, and I'm comfortable operating across cloud infrastructure (AWS, Firebase) as much as I am in the frontend. I'm always learning, always shipping.

## WORK EXPERIENCE

**Senior Software Engineer — Mobile** | **SocialBu** *Aug 2025 – Present (7 months) · Lahore, Pakistan*

Took ownership of SocialBu's mobile app strategy, architecting and delivering cross-platform Android and iOS applications built on a WebView-based foundation.

- Architected and maintained production Android and iOS apps using a WebView architecture, ensuring a consistent experience across both platforms.
- Evaluated, integrated, and experimented with Ionic and Capacitor frameworks to align the mobile layer with evolving product and technical requirements.
- Identified and resolved platform-specific performance bottlenecks on both Android and iOS, significantly improving app stability.
- Collaborated tightly with product and backend teams to keep web and mobile behavior in sync throughout the release cycle.
- Implemented in-app purchase flows using third-party billing services, enabling new monetization capabilities for the product.
- Led release preparation, end-to-end testing, and submission to both the App Store and Google Play.

**Senior Software Engineer — Full Stack** | **SocialBu** *Jun 2024 – Jul 2025 (1 yr 2 mos) · Lahore, Pakistan*

Joined SocialBu to help design and build scalable full-stack solutions across both microservices and monolithic architectures — working primarily on their AI-driven chat product.

- Built an AI Character Chat product end-to-end using Quasar (Vue.js), Firebase Cloud Functions, and Express.js — a platform where users interact with AI personas in natural conversation.
- Integrated ChatGPT and Fireworks AI models to generate contextually rich, human-like chat responses, meaningfully improving user engagement.
- Designed and implemented a microservices architecture using Firebase Cloud Functions and Express.js, enabling the product to scale independently across features.
- Set up Sentry for real-time error monitoring and established structured logging practices, reducing mean time to debug production issues.
- Played an active role in cross-functional decision-making — from ideation and architecture discussions to code reviews and production deployments.
- Partnered with product owners to plan, spec, and prioritize new features; broke down work into detailed technical tasks for the engineering team.

**Software Engineer — Node.js / Backend** | **Tossdown Inc.** *Feb 2023 – Jun 2024 (1 yr 5 mos) · Lahore, Pakistan*

Contributed to a high-volume food-tech platform that processes over 10 million orders annually, taking on both feature development and architectural responsibilities.

- Designed and built RESTful APIs powering the communication layer between multiple frontend clients and the backend at significant scale.

- Implemented secure authentication and authorization systems using JWT, protecting user data across the platform.
- Designed and enforced data validation and encryption practices to maintain privacy compliance and reduce attack surface.
- Set up and maintained CI/CD pipelines using GitHub Actions, reducing deployment time and increasing release confidence.
- Contributed to database design decisions and application architecture discussions to keep the codebase maintainable as the product grew.
- Wrote and maintained thorough API documentation for both internal teams and external integrators.

**Associate Software Engineer — React.js** | **Tossdown Inc.** *Nov 2021 – Jan 2023 (1 yr 3 mos) · Lahore, Pakistan*

- Built reusable, accessible React.js components from UI/UX designs, bridging design and engineering effectively.
- Managed complex application state using Redux Toolkit, Context API, and Redux Saga depending on the use case.
- Optimized React component performance through memoization, lazy loading, and lifecycle method tuning.
- Handled asynchronous data flows (API calls, real-time updates) cleanly to ensure smooth, responsive user experiences.

**Freelance Full-Stack Developer** | **Self-Employed** *Sep 2020 – Nov 2021 (1 yr 3 mos) · Remote*

Worked independently with startups and small businesses to deliver web applications from concept to launch, managing everything from client communication to production deployment.

**Research Assistant** | **Xi'an Key Lab of Wireless Network & Systems** *Oct 2017 – Jul 2019 (1 yr 10 mos) · Xi'an, China*

- Conducted research on Internet traffic prediction, applying Neural Networks, Deep Learning, and Exploratory Data Analysis to large-scale datasets.
- Developed and trained machine learning models to forecast network behavior, contributing findings to lab research initiatives.

## KEY PROJECTS

**SocialBu CLI — Terminal-Based Account Manager** · *Node.js, TypeScript, OpenAPI, REST API, CLI (Commander.js)*

A developer-focused CLI tool that brings the full SocialBu platform to the terminal — no browser required. Built in TypeScript on Node.js and generated from the SocialBu OpenAPI spec, it lets power users and engineers manage posts, accounts, teams, media, and analytics entirely from the command line, with both interactive prompts and scriptable flag-based modes.

- Designed the full CLI architecture from scratch using TypeScript and Node.js, with commands auto-generated from the SocialBu OpenAPI spec — meaning new API endpoints automatically become new CLI commands after running a single generate script.
- Implemented full post lifecycle management — create, update, delete, publish, and calendar view — with interactive prompts for missing inputs and a --no-interactive flag for CI/CD pipeline usage.
- Built secure API key authentication with local config storage, supporting both interactive login flow and environment variable injection for headless and automated environments.
- Delivered human-friendly tabular output by default for all list commands, with --output json for machine-readable output — making it equally usable for humans and scripts.
- Covered the full SocialBu surface area: accounts, teams, media upload, AI content generation, curation, insights/analytics, and notifications — all manageable without opening a browser.

*GitHub: github.com/gMabdullah/socialbu-cli*

**SocialBu — Social Media Manager App (Android & iOS)** · *React Native (WebView), Ionic, Capacitor, Android, iOS, In-App Purchase SDKs*

Live on both the Google Play Store and Apple App Store, SocialBu is a full-featured social media management platform with 10,000+ downloads. The app lets individuals, teams, and businesses manage all their social accounts — Facebook, Instagram, LinkedIn, Twitter/X, and Google My Business — from a single interface. As the lead mobile engineer, I owned the full mobile layer of this product.

- Architected and shipped cross-platform Android and iOS apps using a WebView-based architecture, enabling rapid feature parity with the web product without duplicating the entire codebase.
- Integrated and evaluated Ionic and Capacitor to bridge native device capabilities (camera, notifications, storage) with the web layer, adapting the approach as product requirements evolved.
- Implemented in-app purchase flows for both stores using third-party billing SDKs, unlocking subscription-based monetization for the mobile product.

- Diagnosed and resolved platform-specific rendering and performance issues on Android and iOS, ensuring a consistent, crash-free user experience across device types.
- Managed the full release pipeline — QA, build signing, metadata, and submission — for both Google Play and the Apple App Store across multiple release cycles.

*Google Play: play.google.com/store/apps/details?id=com.socialbu.app · App Store: apps.apple.com/us/app/socialbu-social-media-manager/id1568492955*

**ChatFai — AI Character Chat Platform** · *Quasar (Vue.js), Firebase, Cloud Functions, Express.js, OpenAI, Fireworks AI*

- End-to-end product build: real-time conversational AI platform where users interact with distinct AI personas in a text-based chat experience.
- Integrated OpenAI GPT and Fireworks AI models to produce natural, context-aware responses that feel genuinely human.
- Architected the backend as a microservices system on Firebase Cloud Functions + Express.js for independent scalability per feature.

**PDF Editor — In-Browser Document Editor** · *Vanilla JS, Node.js, Express.js, PDFLib.js, PDF.js*

- Built a browser-based PDF editor with draggable, resizable form fields and rich text formatting (bold, italic, underline, fonts, colors).
- Used PDFLib.js and PDF.js for rendering, editing, and saving documents while preserving field structure and data integrity.

**TikTok Management Tool** · *React.js, Bootstrap, Chart.js*

- Built a social media management dashboard for scheduling TikTok content, tracking analytics (views, likes, shares, follower growth), and managing multiple accounts from one interface.

**Workout App — Serverless Fitness Backend** · *Node.js, AWS Lambda, Serverless Framework, Sequelize ORM, JWT, Google Auth*

- Designed and deployed a serverless REST API on AWS Lambda with Sequelize ORM for workout CRUD, scheduled email notifications, and Google OAuth authentication.

**Ecommerce Admin Panel — Multi-Module Dashboard** · *React.js, TypeScript, Firestore, Chart.js, Material UI, RTK*

- Built a comprehensive admin dashboard covering orders, inventory, loyalty programs, coupons, subscriptions, payment integrations, and real-time analytics — all with dynamic reporting and data visualizations.

## TECHNICAL SKILLS

**Languages & Frameworks:** JavaScript (ES6+), TypeScript, Python, React.js, Vue.js (Quasar), React Native, Node.js, Express.js

**Mobile:** React Native, Ionic, Capacitor, WebView-based Android & iOS apps

**AI / ML Integration:** OpenAI (ChatGPT), Fireworks AI, TensorFlow, Neural Networks, Deep Learning

**Databases:** MongoDB, MySQL, Firestore (Cloud), Sequelize ORM

**Cloud & Infrastructure:** AWS (Lambda, Serverless), Firebase (Cloud Functions, Hosting, Auth), Vercel

**Tools & Practices:** Sentry, Jira/Agile, CI/CD (GitHub Actions), JWT, Docker, Git, Basecamp, Sprint planning

**Other APIs:** Google Maps, Google Auth, Stripe/In-App Purchase, PDFLib.js, Chart.js, Pusher

## EDUCATION

**Master of Science — Computer Science & Technology** *2017 – 2019*

*Xi'an Jiaotong University, Xi'an, China · Focus: Deep Learning & Machine Learning · Chinese Government CSC Fully Funded Scholarship*

**Bachelor of Science — Information Technology (Web Development)** *2010 – 2014*

*University of the Punjab, Lahore, Pakistan · University Scholarship Recipient · Prime Minister Laptop Scheme Recipient*

## CERTIFICATIONS & HONORS

- Machine Learning — Coursera / Stanford
- Microsoft Unlimited Potential — Web Development
- Excel Skills for Data Analytics and Visualization
- Chinese Government Scholarship Council (CSC) — Fully Funded Graduate Scholarship
- Prime Minister of Pakistan Laptop Scheme Awardee
- University Merit Scholarship — University of the Punjab